GEOGRAPHY

PHYSICAL GEOGRAPHY

CLIMATOLOGY

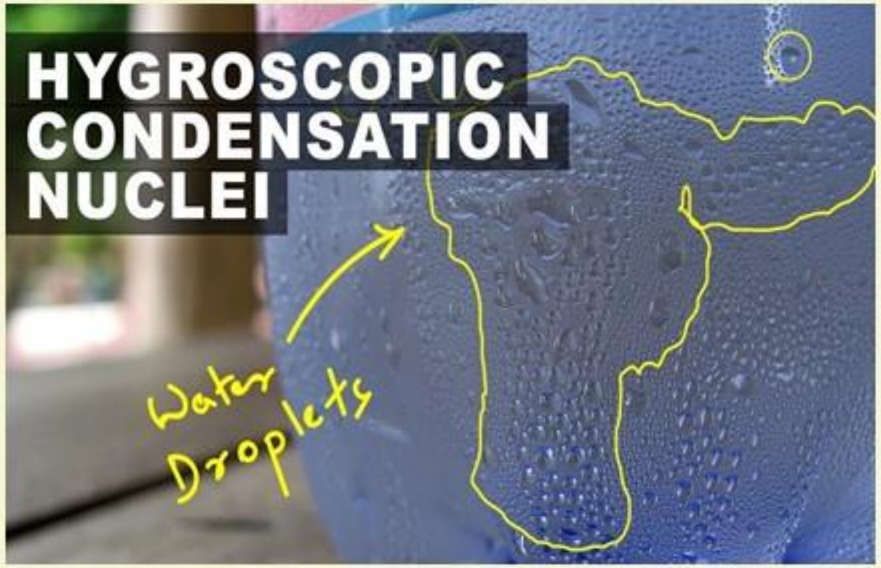
MODULE – 27 | CLOUDS

CLOUDS

1. Clouds consist of very small droplets of water or minute crystals of ice, so small that they can be sustained by even the slightest upward movement of air.
2. It is imperative that condensation takes place for clouds to form and water vapour is converted into water or ice particles.

CLOUDS

PROCESS OF CLOUD FORMTION

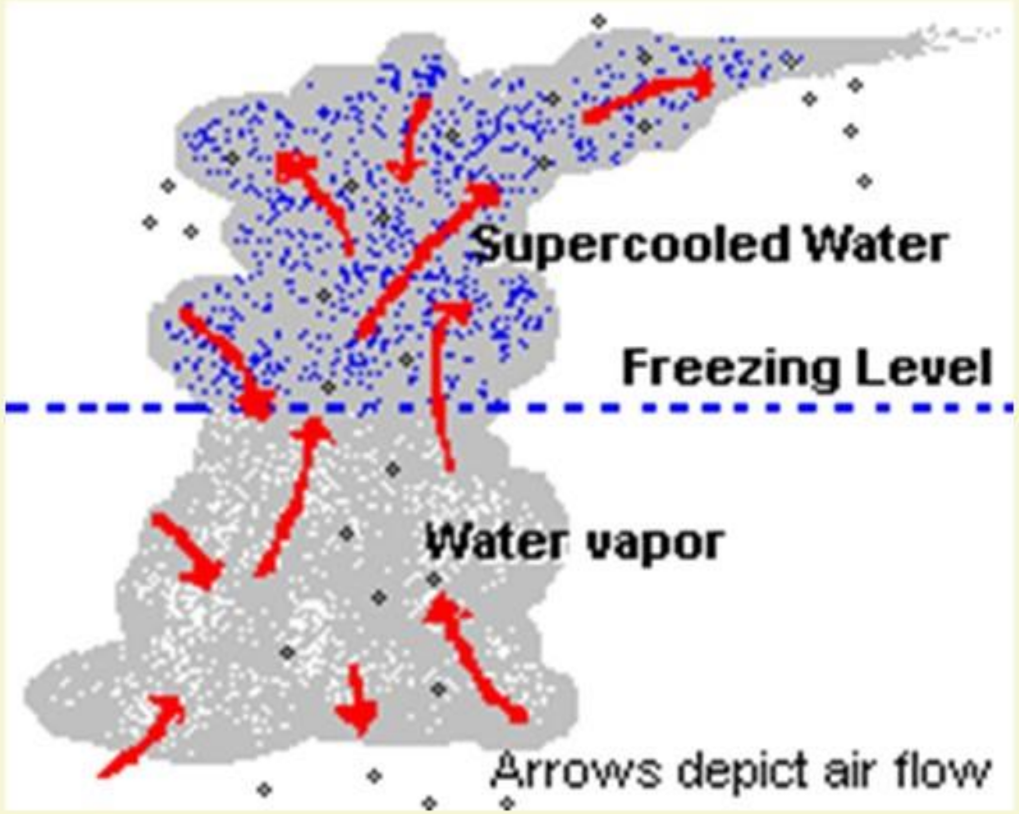


1. In order for cloud droplets to form it is also necessary that microscopic particles of dust or smoke with a high affinity for water (hygroscopic particles) are present in air and serve as condensation nuclei.

2. In the absence of hygroscopic particles in the atmosphere, condensation becomes difficult even though the air may have cooled below the dew point.

CLOUDS

PROCESS OF CLOUD FORMTION



3 Water vapour in the air that fails to condense due to lack of hygroscopic particles or any other cause in spite of the air parcel having been cooled below the dew point is called super-cooled moisture.

4 This phenomenon is rare and generally confined to the upper parts of the troposphere.

CLOUDS

TYPES OF CLOUDS

1. Clouds are of different types and they can be classified on the basis of their form and altitude.

2. On the basis of form clouds are known as
   1. Stratiform or layered types
   2. Cumuliform or massive types

STRATIFORM CLOUDS

CUMULIFORM CLOUDS

CLOUDS

TYPES OF CLOUDS

NIMBUS CLOUDS

3 On the basis of altitude, clouds are called
1. cirrus (highest clouds) and
2. alto (medium height)
3. Strato (low-level clouds)

4 The combination of these two lead to the cirrocumulus, cirrostratus, altocumulus and altostratus types.

5 To identify a cloud from which rain or snow falls, the term nimbo or nimbus is appended to the cloud types, e.g. nimbostratus and cumulonimbus.

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

1 HIGH CLOUDS

a CIRRUS CLOUDS

CIRRUS CLOUDS

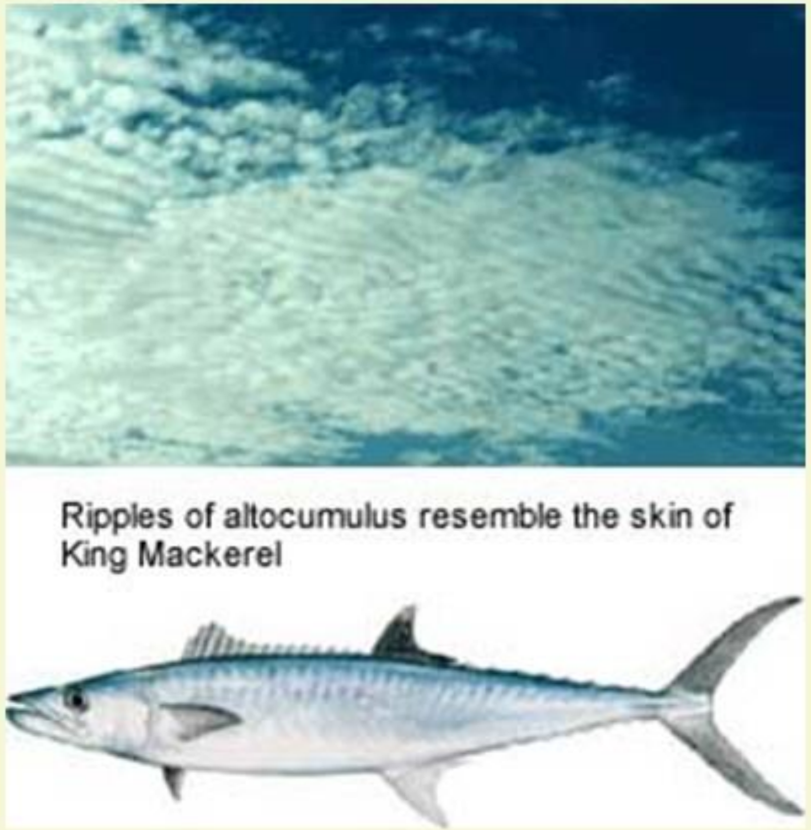
1. They occur at altitudes of 6000 to 12000 metres and include cirrus, cirrocumulus and cirrostratus clouds.
2. Cirrus is a wispy, fibrous cloud and is associated with fair weather.
3. They are composed of ice crystals that originate from the freezing of supercooled water droplets.

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

1 HIGH CLOUDS

b CIRROCUMULUS CLOUDS



Cirrocumulus is a thin, often globular cloud. This shows ripples and presents the mackerel sky conditions.

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

1 HIGH CLOUDS

c CIRROSTRATUS CLOUDS

CIRROSTRATUS CLOUDS

Cirrostratus clouds look like a thin, light coloured, often white sheet. The sun and moon produce a halo through this type of clouds.

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

2 MEDIUM CLOUDS

a ALTOSTRATUS CLOUDS

ALTOSTRATUS CLOUDS

1. They form at an altitude of 2100 to 6000 metres above the surface and are divided into the categories of altostratus and the altocumulus clouds.
2. The altostratus cloud produces a blanket layer covering vast areas of the sky.
3. They have a smooth underside and the sun is visible as a bright spot through them.
4. Grayish in colour, they are associated with the development of bad weather.

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

2 MEDIUM CLOUDS

b ALTOCUMULUS CLOUDS

ALTOCUMULUS CLOUDS

1. The altocumulus clouds present geometrical patterns produced through fitting together of layers of individual cloud masses.

2. The clouds of this type are often white or gray and patches of blue sky are visible. The clouds look bumpy, have a flattened base and indicate fair weather.

Learning Space

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

3 LOW CLOUDS

b STRATUS CLOUDS

STRATUS CLOUDS

1. This group includes clouds occupying heights of up to 2,100 metres. This group includes the stratus, the nimbostratus and the stratocumulus types.
2. Stratus is a low dense cloud looking like fog and presenting a dark grey look. Associated with bad weather, it usually brings drizzle. Fog is very low stratus cloud.

Learning Space

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

3 LOW CLOUDS

b STRATOCUMULUS CLOUDS

STRATOCUMULUS CLOUDS

1. Stratocumulus is also a low cloud. It consists of distinctly grayish masses of cloud between which blue patches of sky are visible.
2. Generally oriented at right angles to the wind direction, such clouds are indicators of fair or clear weather.

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

3 LOW CLOUDS

c NIMBOSTRATUS CLOUDS

NIMBOSTRATUS CLOUDS

1. If the stratus cloud is causing rainfall, It is called nimbostratus.

CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

CUMULIFORM CLOUDS

a CUMULUS CLOUDS

CUMULUS CLOUDS

1. As mentioned earlier, they are massive clouds having a vertical extent from 1500 to 9000 metres. They have a low, flat base and appear as isolated cloud masses.
2. When sunlit, these clouds appear shining white and are called wool clouds.
3. Produced by convection, they are characteristic during the summer season. Small cumulus clouds indicate a fair weather.

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CLOUDS

CHARACTERISTICS OF THE MAJOR TYPES OF CLOUDS

CUMULIFORM CLOUDS

b CUMULONIMBUS CLOUDS

CUMULONIMBUS CLOUDS

1. They are dense towering vertical cloud, with an anvil shape, associated with thunderstorms and atmospheric instability.
2. It has a huge size and causes heavy precipitation accompanied by thunder and lightening.
3. If seen from a distance, the top of the cloud may appear white but immediately below it, the daylight may be totally obscured and the cloud appears almost black.

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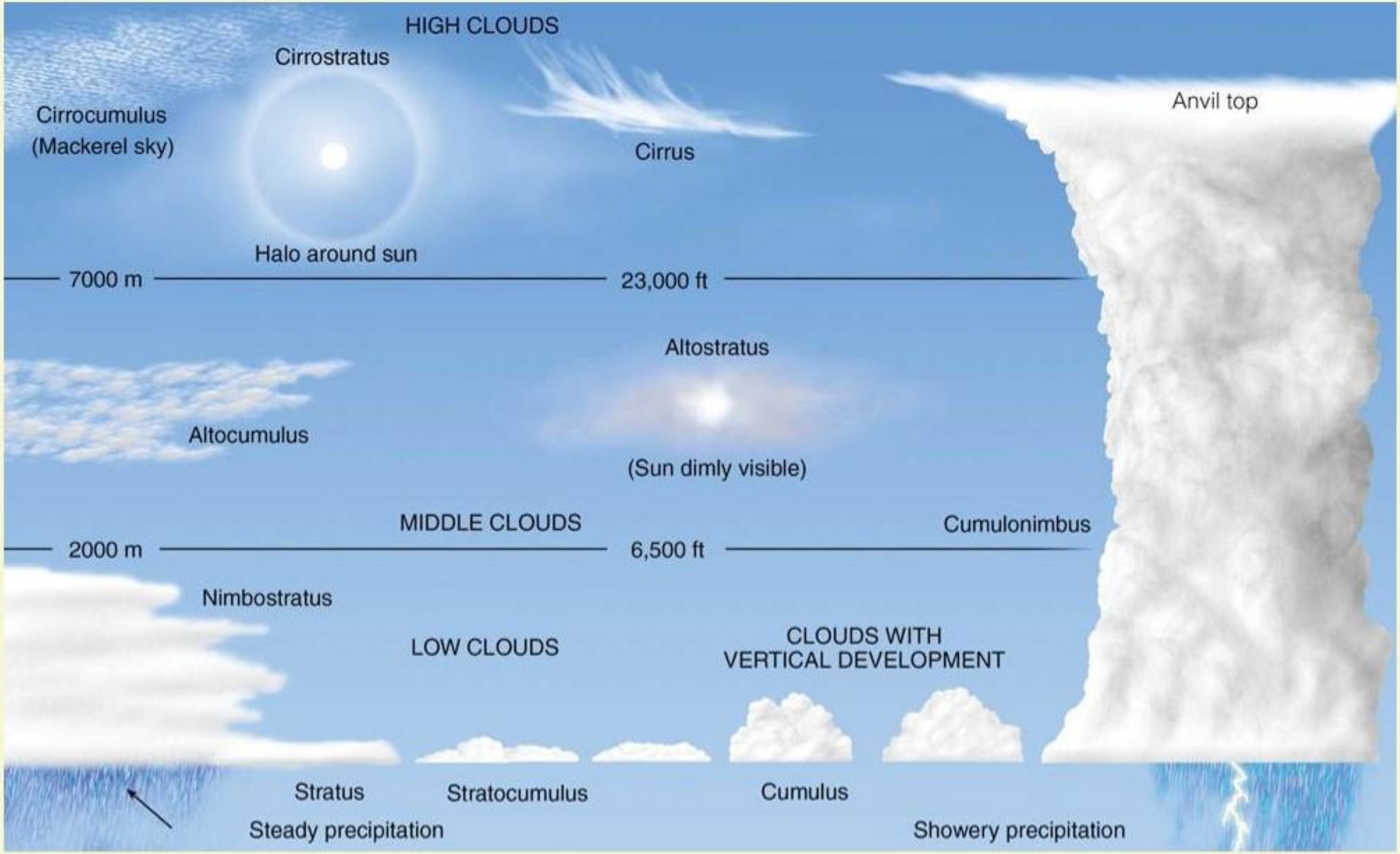
To sum it up, the various clouds are...



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UPSC CIVIL SERVICES

PRELIMS - 2018

MCQs DISCUSSION

GEOGRAPHY

SESSION - 11

31st May, 2018

1 11 NEW CLOUD TYPES ADDED TO THE INTERNATIONAL CLOUD ATLAS

WHAT IS THE INTERNATIONAL CLOUD ATLAS?

- The International Cloud Atlas was first created in 1896 and has been a resource of cloud types and photos that has helped train meteorologists for decades.
- 2017 was the year where the first digitized/online version of the Atlas for WMO was released.

1 11 NEW CLOUD TYPES ADDED TO THE INTERNATIONAL CLOUD ATLAS

WORLD
METEOROLOGICAL
ORGANIZATION

- The increased use of technology is what prompted the World Meteorological Organization to add 11 new cloud classifications to their International Cloud Atlas, a globally recognized source for meteorologists.
- These 11 additions are the first updates that the atlas has received in 30 years, and much of the change can be attributed to citizen scientists who can share and discuss clouds by uploading photos to the Atlas's site.

1 11 NEW CLOUD TYPES ADDED TO THE INTERNATIONAL CLOUD ATLAS

ASPERITAS

- Asperitas, Latin for roughness, is the cloud type that has citizen scientists most excited and has been a special victory for the UK-based Cloud Appreciation Society.
- Marked by small divot-like features that create chaotic ripples across the sky, asperitas were championed by enthusiasts who noticed they did not accurately fall under existing categories.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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Learning Space

For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

OUR TEAM

G. V. Rao	Amrita Naidu
K. Srikanth	M. Binukrishna
D. Sunil Kumar	G. Ravi Babu
L. V. Krishna	D. Joji
D. Chaitanya	K. Victor Babu

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